

Nathan Tang

nathan.ly.tang@gmail.com | +1 (780)-908-8238 | nathanlytang.com | github.com/nathanlytang
Canadian citizen eligible for U.S. employment with TN status, open to relocation.

EXPERIENCE

People Inc. (Dotdash Meredith) *Software Engineer*

May 2022 – Present

- Co-architected a **full-stack** ad configuration and delivery platform (API, database, management UI), enabling instant global updates across the ad network, contributing **\$1M/day in revenue**.
- Led architectural decisions for backend services, scaling to handle **millions of daily API calls**, including database schema design, data sorting logic, and authentication systems.
- Developed a web-based configuration CMS with OAuth-secured access using modern frameworks.
- Built reusable components adopted across all brands, enabling flexible configuration and improving viewability while sustaining strong Core Web Vitals scores.
- Led optimization of backend data processing for dynamic bundling, recursive configuration sorting, and configuration tiering, **reducing payload size by 30%** and improving API response times.
- Built a deduplication and tiering system that merged overlapping ad configurations, **reducing database footprint by 40%** and server load substantially.
- Built tooling to automate API schema-to-DTO generation, creating strongly typed backend–frontend interfaces, speeding up development, and reducing runtime errors through consistent contracts.

Aquatic Biosphere Society of Canada *Full Stack Engineer*

January 2024 – April 2024

- Created and shipped an educational mobile augmented reality game in partnership with the University of Alberta, showcased live at the Edmonton Boat and Sportsmen's Show, within 3 months.
- Developed a **computer vision pipeline** using OpenCV for mural image recognition on mobile, optimized for fast processing under real-world bandwidth and device constraints.
- Built a custom web **admin CMS** interface for content teams to manage game types, game content, facts, scores, and user data with real-time updates and live content editing.
- Designed core APIs** for gameplay mechanics, scoring, user tracking, and CMS functionality, integrating both the game and admin tools with Google OAuth authentication.
- Built the backend infrastructure, leveraging Azure's web application and database services.

Snap & Tow *Backend Engineering Intern*

March 2022 - May 2022

- Designed and built the backend system and database schema for a real-time roadside assistance platform.
- Integrated with **Google Maps** API to match users with the closest operators in real time based on location.

PROJECTS

thechoice.fm | PHP, Javascript, Node.js, MySQL

- Developing a real-time on-demand web radio application enabling synchronized playback for all users via a distributed playlist system, with a shared queue experience.
- Building an API and admin panel to manage song queues, interfacing with radio automation software.

Discord Bot | TypeScript, Node.js, Docker, MySQL

- Developed a bot that queries game servers in real time using the UDP Query protocol to return live status and player information. Designed for resiliency using Docker for isolation and persistent state.
- Deployed to 700+** Discord communities and serves **hundreds of active users** daily.

EDUCATION

University of Alberta | Bachelor of Science in Computing Science

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, Go (Golang), Java, C, HTML/CSS, SQL
Databases	MySQL, PostgreSQL, MongoDB, Redis, Firebase
Technologies	Node.js, React, Vue, Svelte, Tailwind, Express.js, Django, Git, Docker, AWS, Azure